Full Radial Numerical Stability of the Solved DTFE Hedgehog Branch
Scale Nondegeneracy and Radial Jacobi Positivity

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Abstract

This paper records a standalone numerical stability result for the canonical hedgehog branch of the Darke Theory Field Equation frozen matter sector. The claim is intentionally narrow. The paper does not assert a complete infinite-domain stability theorem, nor does it claim that every remaining analytic obligation in the broader DTFE programme has been discharged. Its contribution is more specific. The actual reduced hedgehog branch is solved numerically by shooting, its virial balance is verified to high accuracy, the reduced scale quotient is evaluated directly on the solved branch and found strictly positive on a certified interval, and the full radial hedgehog Jacobi operator is discretised on multiple grids with strictly positive lowest Dirichlet eigenvalue throughout the tested ladder. In addition, the regular Jacobi shooting derivative does not enter a decaying zero-mode class, yielding a nonzero transversality signal at large radius.

The central numerical question of the paper is whether the solved hedgehog branch remains stable under the most comprehensive full-radial numerical assessment currently implemented in the DTFE engine. The answer is affirmative. The branch is numerically realised with small tail error, near-unit topological charge, and tight virial residual. The reduced scale curvature is strictly positive, the full radial Jacobi spectra remain positive on the tested grids, and the transversality obstruction is nonzero. These results do not yet constitute a proof on the infinite half-line, but they replace earlier ansatz-level approximations and finite basis surrogates with a direct numerical analysis of the solved branch itself. The paper therefore establishes a substantive numerical stability result for the radial DTFE hedgehog sector.

1 Introduction

The DTFE frozen-sector programme has reached a stage at which numerical evidence must be stated with precision and restraint. Earlier computational investigations provided encouraging but structurally limited results: a virial-optimised ansatz branch, positive reduced scale curvature on that profile family, and positive behaviour in a finite basis surrogate for the $K = 1$ sector. Those results were informative, but they did not meet the stronger numerical burden required for a standalone stability paper. The most natural objection remained clear. One had not yet solved the actual reduced hedgehog branch, and one had not yet subjected that solved branch to a full radial Jacobi analysis.

The present paper addresses exactly that objection.

Its purpose is not to provide a broad account of DTFE, nor to rehearse the full theorem programme. It has one narrow goal:

> *to determine whether the solved hedgehog branch of the DTFE frozen sector remains numerically stable under a consolidated full-radial assessment directed at scale degeneracy and radial Jacobi instability.*

That question breaks naturally into four components.

(i) Does the actual reduced hedgehog branch exist numerically under the correct boundary conditions?

(ii) Is the realised branch genuinely close to virial balance?

(iii) Is the reduced scale quotient nondegenerate on the solved branch itself?

(iv) Does the full radial hedgehog Jacobi operator exhibit an immediate negative mode or a zero-mode transversality collapse?

The present paper answers all four in the numerically favourable direction.

The solver uses a genuine shooting construction for the hedgehog boundary value problem rather than an optimised ansatz family. The realised branch is then audited through its energy split, virial residual, tail value, topological charge, and characteristic radii. The scale quotient is computed directly on the solved branch and enclosed over a finite interval around physical scale. The full radial Jacobi operator is assembled on multiple radial grids and its lowest Dirichlet eigenvalues are computed. Finally, the regular Jacobi shooting derivative is propagated to large radius and tested for transversality, so that numerical positivity of finite-window spectra is not confused with an unresolved near-zero mode.

The result is the first DTFE numerical paper in this sequence whose evidential basis is centered on the solved branch itself rather than on reduced surrogate structure. The result remains numerical and finite-window in character. But it no longer depends on ansatz-level evidence alone.

2 Exact claim boundary

The value of the paper depends on the exactness of its claim boundary. We therefore state explicitly what is and is not being asserted.

2.1 What is claimed

Proposition 2.1 (Standalone numerical claim boundary)**.** *Within the presently implemented DTFE frozen-sector engine:*

(a) the reduced hedgehog branch is numerically realised by shooting;

(b) the realised branch is close to virial balance and close to unit topological charge;

(c) the reduced physical scale direction is strictly nondegenerate on a certified interval about the solved branch scale;

(d) the full radial hedgehog Jacobi operator has strictly positive lowest Dirichlet eigenvalue on each tested grid in the fixed grid ladder;

(e) the regular Jacobi shooting derivative has nonzero transversality at large radius.

2.2 What is not claimed

The paper does *not* claim:

(a) an infinite-domain proof of radial positivity;

(b) a final analytic classification of the full Jacobi operator on $(0, \infty)$;

(c) completion of every theorem-level obligation in the full DTFE stability programme;

(d) a proof of nonradial stability beyond the present radial analysis.

Remark 2.2. *This distinction is essential. A serious numerical paper loses its value immediately if it presents finite discretisation results as though they were already infinite-domain theorems.*

3 The fixed DTFE sector used in this paper

The present paper does not rederive the entire DTFE frozen sector. It treats the frozen matter sector, its hedgehog reduction, and its energy decomposition as fixed background from the broader DTFE programme. Only the ingredients required for the present numerical analysis are recorded here.

3.1 Hedgehog branch

We work with the canonical hedgehog profile $F(r)$ on the radial half-line, subject to the standard boundary conditions

$$F(0) = \pi, \qquad F(\infty) = 0. \tag{1}$$

The realised numerical branch analysed in this paper is the branch denoted `hedgehog_B1`.

3.2 Energy split

The total frozen-sector energy is decomposed as

$$E = E_2 + E_4, \tag{2}$$

where E_2 and E_4 denote the standard lower-order and higher-order contributions used by the DTFE frozen-sector engine. The present paper uses this split only to define and audit virial balance numerically.

3.3 Scale quotient

The scale direction is tested by physically rescaling the solved hedgehog branch and computing the second derivative of total energy with respect to the scaling parameter λ. The reduced scale obstruction is therefore

$$\mathcal{O}_{\text{scale}} = \left.\frac{\mathrm{d}^2 E(\lambda)}{\mathrm{d}\lambda^2}\right|_{\lambda=1}, \tag{3}$$

computed numerically by a five-point stencil and enclosed on a finite interval about $\lambda = 1$.

3.4 Full radial Jacobi operator

The radial Jacobi analysis is built from the solved branch itself. In the current implementation the full radial operator is recorded in the form

$$J_F \eta = -2 \frac{\mathrm{d}}{\mathrm{d}r}\Big((r^2 + 2\sin^2 F)\eta' \Big) + V_{\text{eff}}(r)\eta, \tag{4}$$

with regular-origin and decaying-tail conditions approximated on finite radial windows. The closed form of V_{eff} is not needed in the present paper beyond the fact that it is assembled directly from the solved hedgehog branch rather than from an external ansatz family.

4 Numerical protocol

4.1 Branch solver

The branch is solved by a shooting boundary-value construction using an RK4 radial integrator. The solver mode reported by the engine is

`shooting_bvp_rk4`.

The shooting parameter is adjusted until the large-radius profile reaches the target boundary class with small tail error.

4.2 Branch audit

Once the branch is realised, the engine records:

(a) total energy E and energy split (E_2, E_4);

(b) virial residual;

(c) topological charge;

(d) characteristic radii $r_{1/2}$ and r_{rms};

(e) shooting parameter and tail value.

4.3 Scale analysis

The scale analysis has two layers:

(a) a point estimate of $\mathcal{O}_{\mathrm{scale}}$ by five-point second derivative on the solved branch;

(b) an interval-subdivision Lipschitz sign check on a fixed scale window about $\lambda = 1$.

4.4 Jacobi analysis

The full radial Jacobi operator is discretised on the fixed grid ladder

$$N_r \in \{240, 360, 480\}, \tag{5}$$

and the lowest Dirichlet eigenvalue is computed on each grid. The minimum of these lowest eigenvalues is recorded as the Jacobi obstruction reported by the engine.

4.5 Transversality analysis

A regular Jacobi shooting derivative is propagated to large radius. The engine records:

(a) the transversality radius;

(b) the transversality value;

(c) an asymptotic $A(\infty)$ estimate;

(d) the finite difference step used for the shooting derivative.

The requirement imposed here is nonvanishing. The analysis is not intended to supply a complete asymptotic theory. It asks only whether the numerically generated variation appears to collapse into a decaying zero-mode class. It does not.

Table 1: Solved hedgehog branch summary

Quantity	Value
Branch ID	`hedgehog_B1`
Solver mode	`shooting_bvp_rk4`
Grid points	2400
Radial interval	$[10^{-4}, 30]$
Boundary condition	$F(0) = \pi,\ F(\infty) = 0$
Shooting parameter	2.0079694325114694
Total energy	11.607477832874274
E_2	5.8026948171598605
E_4	5.804783015714413
Virial residual	$1.7990114515995626 \times 10^{-4}$
Topological charge	1.0004796463783436
$r_{1/2}$	0.9379876615256356
r_{rms}	1.4371514542259483
Tail value	-0.0017076504814172414

5 Realised branch data

The solved branch data are taken directly from the current engine output.

Remark 5.1. *The realised branch is numerically clean in every obvious respect relevant to the present analysis. The tail is small, the topological charge is close to unity, and the virial residual is sufficiently small that the branch cannot reasonably be dismissed as a loose or visibly unbalanced solution profile.*

6 Virial balance

The branch satisfies

$$E_2 = 5.8026948171598605, \qquad E_4 = 5.804783015714413, \tag{6}$$

so that the split is extremely close to the virial condition $E_2 \approx E_4$. The engine records the virial residual as

$$\mathrm{VirialResidual} = 1.7990114515995626 \times 10^{-4}. \tag{7}$$

Proposition 6.1 (Virial tolerance)**.** *The realised branch satisfies the virial tolerance used in the consolidated numerical assessment.*

Proof. The virial tolerance requires the virial residual to remain below the fixed numerical cutoff used by the engine. The realised residual lies comfortably below that threshold. □

7 Scale nondegeneracy on the solved branch

The reduced scale obstruction recorded by the engine is

$$\mathcal{O}_{\mathrm{scale}} = 11.034283298811214. \tag{8}$$

This value is already strongly positive. The more important point, however, is interval sign stability rather than the point estimate alone.

The certified scale interval lower and upper bounds are

$$\mathcal{O}_{\text{scale}}^{\text{lower}} = 9.898580118938215, \qquad \mathcal{O}_{\text{scale}}^{\text{upper}} = 13.707750844046373. \tag{9}$$

The tested scale interval is the fixed window

$$\lambda \in [0.95, 1.05], \tag{10}$$

subdivided into twelve cells, each of which retains positive certified lower bound.

Theorem 7.1 (Numerical scale nondegeneracy)**.** *Within the tested scale window, the solved DTFE hedgehog branch is numerically nondegenerate in the reduced scale quotient.*

Proof. Every subdivision cell in the certified scale interval has positive lower bound. Therefore no sign change of the reduced scale curvature is detected anywhere on the tested interval. The solved branch is therefore numerically nondegenerate under physical scaling in that window. □

Remark 7.2. *This is materially stronger than the earlier ansatz-family evidence. The current obstruction is computed on the solved shooting branch itself.*

8 Full radial Jacobi analysis

8.1 Lowest eigenvalues

The lowest Dirichlet eigenvalues of the full radial Jacobi operator on the tested grid ladder are

$$\lambda_{\min}^{(240)} = 3.3169113879177616, \tag{11}$$

$$\lambda_{\min}^{(360)} = 3.8408370066762263, \tag{12}$$

$$\lambda_{\min}^{(480)} = 3.3340862302193757. \tag{13}$$

All three are strictly positive.

The minimum of these values, used by the engine as the Jacobi obstruction, is

$$\mathcal{O}_{\text{Jacobi}} = 3.3169113879177616. \tag{14}$$

8.2 Grid spread

The final report records grid relative spread

$$\text{GridSpreadRel} = 0.1498095340311708. \tag{15}$$

Thus the positivity signal is not produced by a single isolated grid or by an obviously unstable discretisation point. The values move across the ladder, but remain consistently positive.

Theorem 8.1 (Full radial numerical Jacobi positivity on the tested grid ladder)**.** *The solved DTFE hedgehog branch exhibits strictly positive lowest Dirichlet eigenvalue for the full radial Jacobi operator on each tested grid in the fixed ladder* $\{240, 360, 480\}$*.*

Proof. This follows directly from the recorded spectra. The lowest eigenvalue on each tested grid is strictly positive. □

Remark 8.2. *This theorem is numerical and finite-ladder in character. It is not an infinite-domain proof. Its importance is that it removes the earlier objection that only finite basis surrogates had been examined.*

9 Transversality and zero-mode obstruction

The transversality data reported by the engine are

$$r_{\text{trans}} = 27.01126465193831, \tag{16}$$
$$\mathcal{T} = -24.17240142258585, \tag{17}$$
$$A(\infty)_{\text{est}} = -0.2983002552708598, \tag{18}$$
$$h_{\text{shoot}} = 5 \times 10^{-5}. \tag{19}$$

The criterion imposed by the final report is simply that the transversality value be nonzero. That condition is satisfied decisively.

Proposition 9.1 (Nonzero transversality)**.** *The present Jacobi shooting derivative does not numerically collapse into a decaying zero-mode class.*

Proof. The transversality value at the recorded large radius is nonzero and of substantial magnitude. The associated asymptotic amplitude estimate is also nonzero. The numerical variation therefore does not enter a decaying zero-mode class under the present analysis. □

Remark 9.2. *This is an obstruction statement, not a complete asymptotic theorem. Its purpose is to exclude the obvious numerical possibility that positivity of finite-window spectra is concealing an unresolved near-zero transversality failure.*

10 Consolidated numerical stability assessment

For the purposes of the present paper, the radial numerical evidence is aggregated into a fixed stability assessment with the following criteria:

(1) existence of the solved branch;

(2) satisfaction of the virial tolerance;

(3) satisfaction of the tail tolerance;

(4) satisfaction of the topological-charge tolerance;

(5) positivity of the reduced scale obstruction;

(6) positivity of the certified lower bound on the scale interval;

(7) positivity of the lowest radial Jacobi eigenvalue on each tested grid;

(8) consistency of the eigenvalue ladder within the prescribed spread tolerance;

(9) nonvanishing transversality.

The recorded output satisfies all nine criteria.

Theorem 10.1 (Consolidated radial numerical stability result)**.** *The solved DTFE hedgehog branch satisfies all criteria of the present consolidated full-radial numerical stability assessment.*

Proof. Direct from the recorded branch, scale, Jacobi, and transversality data. Each criterion listed above is satisfied by the current numerical output. □

Table 2: Consolidated gate summary

Criterion	Result
Existence of solved branch	satisfied
Virial tolerance	satisfied
Tail tolerance	satisfied
Topological-charge tolerance	satisfied
Scale obstruction positivity	satisfied
Scale-interval lower-bound positivity	satisfied
Jacobi positivity on all tested grids	satisfied
Grid-consistency tolerance	satisfied
Transversality nonvanishing	satisfied

11 What the result means

The present result establishes a numerically substantive fact about the solved DTFE hedgehog branch. The realised branch simultaneously satisfies branch existence, small virial defect, small tail defect, near-unit topological charge, positive reduced scale curvature on the solved profile, positive certified lower bound on the tested scale interval, positive lowest full-radial Jacobi eigenvalue across the tested grid ladder, and nonzero transversality.

That combination is sufficient to justify a standalone numerical stability paper. It is not sufficient to justify a final analytic stability theorem. It is, however, sufficient to state in a precise and defensible way that the solved branch remains numerically stable under the present full-radial assessment.

12 What remains open

The remaining tasks are now much more sharply defined than before.

12.1 Infinite-domain radial positivity

The current Jacobi result is obtained on finite radial windows with Dirichlet truncation. What remains is an infinite-domain enclosure or a comparison theorem strong enough to convert the present spectra into a rigorous positivity result on the half-line.

12.2 Analytic transversality

The present transversality signal is numerically decisive, but remains numerical. One would still want an analytic or validated-ODE statement showing that the relevant Jacobi derivative cannot enter the decaying zero-mode class.

12.3 Beyond radial

The current paper is radial. It does not settle the full nonradial spectrum.

Remark 12.1. *This is now the correct location of the remaining burden. The principal achievement of the present paper is that it relocates the uncertainty to the appropriate analytic questions. The difficulty is no longer that the numerical evidence is confined to ansatz-level approximations or finite basis surrogates. It is now concentrated where it belongs: infinite-domain closure and nonradial extension.*

13 Conclusion

This paper records the first DTFE stability result in this sequence whose evidential basis is centered directly on the solved hedgehog branch. The actual reduced hedgehog branch has been solved numerically by shooting under the correct boundary conditions. The branch is close to virial balance, close to unit topological charge, and has small tail error. The reduced scale quotient is strictly positive and remains positive on a certified interval about physical scale. The full radial hedgehog Jacobi operator has strictly positive lowest Dirichlet eigenvalue across the tested grid ladder. The regular Jacobi shooting derivative has nonzero transversality at large radius and does not numerically collapse into a decaying zero-mode class. The solved branch therefore satisfies the present consolidated radial numerical stability assessment.

This does not yet establish the infinite-domain theorem. It does, however, establish an important numerical result in a precise and defensible form. The solved DTFE hedgehog branch satisfies the present full-radial numerical stability assessment implemented on the realised profile itself. That is sufficient to move the discussion beyond preliminary reduced-profile evidence and into a properly theorem-facing radial stability programme.

A Numerical report record

For reproducibility, the current run is defined by the following engine report values:

- solver mode: `shooting_bvp_rk4`;
- branch ID: `hedgehog_B1`;
- shooting parameter: 2.0079694325114694;
- tail value: -0.0017076504814172414;
- total energy: 11.607477832874274;
- virial residual: $1.7990114515995626 \times 10^{-4}$;
- topological charge: 1.0004796463783436;
- scale obstruction: 11.034283298811214;
- scale interval lower bound: 9.898580118938215;
- Jacobi lowest eigenvalues: 3.3169113879177616, 3.8408370066762263, 3.3340862302193757;
- grid relative spread: 0.1498095340311708;
- transversality value: -24.17240142258585.

B Suggested status line

A concise status line suitable for repository indexing, front matter, or archive metadata is:

> *Standalone numerical stability paper. Solved DTFE hedgehog branch satisfies a consolidated full-radial numerical stability assessment. Not yet an infinite-domain proof.*